



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: BCA
Semester: VI
Name of the Paper: Introduction to ML
Paper Code: TBC 601
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

- Q1. (10 Marks)
a) Define Mean, Median, and Mode. Explain their significance in machine learning and Calculate Mean, Median, and Mode for the data: 12, 15, 18, 20, 22, 22, 25, 30 (CO1)
OR
b) Explain Range, Absolute Deviation, and Average Deviation. Find Range and Average Deviation for: 10, 12, 15, 18, 20 (CO1)
- Q2. (10 Marks)
a. Explain the basic concepts of Probability Theory. Define sample space, event, and conditional probability with examples. (CO1)
OR
b. Explain different matrix operations such as Addition, Subtraction, scalar Multiplication, each with a suitable example. (CO1)
- Q3. (10 Marks)
a. What is Machine Learning? Explain its need and importance in modern computing systems. (CO1)
OR
b. Explain Supervised Learning in detail. Discuss its working principle and give suitable real-world examples. (CO1)
- Q4. (10 Marks)
a. Explain Mutable vs Immutable data types in Python Programming with examples (CO2)
OR
b. Discuss file modes and explain read, write, and append operations with examples. (CO2)
- Q5. (10 Marks)
a. Explain Built-in functions & User-defined functions in Python with suitable examples (CO2)
OR
b. Explain the following (a) if-elif- else loop and (b) Nested loops (CO2)